

JVM Lab: JVM GC Allocation Rate Tuning

Duration : 60 Min.

Scenario

An application is experiencing high GC allocation rates, causing premature object promotion and frequent Full GC cycles.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Size `G1RegionSize` and `G1ReservePercent` boundaries, reduce allocation rate, and validate heap stability using performance logs.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. `G1RegionSize` configuration tune-up: 5 marks
2. `G1ReservePercent` optimization for promotion rates reduction: 5 marks
3. Young generation allocation benchmark validation metrics: 5 marks
4. Load test diagnostics report comparison logs: 5 marks

Total Score: 20 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.